

Deployment Report — Site & Inventory Manager

A short record of how and where this app is hosted, so anyone picking this up later doesn't have to reverse-engineer it.

Live URLs

Layer	Provider	URL
Frontend	Vercel (free)	https://site-inventory-manager.vercel.app
Backend API	Render (free)	https://site-inventory-backend-mc1y.onrender.com
Database	Neon Postgres (free)	managed via the Neon dashboard
Source code	GitHub	https://github.com/Roshan-Khatiwada-Official/site-inventory-manager

Architecture

Browser → Vercel (static React/Vite build)
 ■ fetch() calls, with x-api-token header
 ▼
 Render (Node/Express API, auto-deploys from `main`)
 ■ Prisma ORM
 ▼
 Neon (managed Postgres)

- **Frontend** (`frontend/`): built with `npm run build`, served as static files by Vercel's CDN. Config: `frontend/vercel.json`. Root directory on Vercel is set to `frontend`.
- **Backend** (`backend/`): Node + Express + Prisma. Deployed from `render.yaml` at the repo root (a Render "Blueprint" — Render reads this file to know how to build/start the service). Root directory on Render is `backend`.
- **Database**: Neon's free tier (0.5 GB storage). The schema and all existing data (users, sites, inventory, assignments, logs) were migrated from the local development database at setup time.

Auto-deploy

Both Vercel and Render are connected directly to the `main` branch of this GitHub repo. Every push to `main` triggers a new build and deploy on both sides automatically — no manual redeploy step needed after a code change.

Environment variables

Actual secret values live only in the Render/Vercel dashboards, never in this repo. Names and purpose, for reference:

Render (backend) → Environment tab:

Variable	Purpose
<code>DATABASE_URL</code>	Neon connection string (pooled)
<code>CORS_ORIGIN</code>	Locked to https://site-inventory-manager.vercel.app — only this origin may call the API from a browser
<code>API_TOKEN</code>	Shared-secret header the frontend must send on every request

Vercel (frontend) → Project Settings → Environment Variables:

Variable	Purpose
<code>VITE_API_URL</code>	Points at the Render backend URL above
<code>VITE_API_TOKEN</code>	Must match <code>API_TOKEN</code> on Render exactly

To rotate the API token: change `API_TOKEN` on Render and `VITE_API_TOKEN` on Vercel to the same new value, save both (each redeploys on save).

Known limitation: cold starts

The backend is on Render's **free** tier, which sleeps after ~15 minutes of no traffic. The first request after that wakes it up, taking roughly 10–50 seconds before the app responds. Every request after that is normal speed until it goes idle again.

To remove this entirely, upgrade only the Render service to the Starter plan (~\$7/month) — the frontend (Vercel) and database (Neon) can stay free either way.

Free-tier limits to watch

- **Neon:** 0.5 GB database storage (current usage is well under this).

- **Render free web service:** sleeps when idle (see above); no storage limit concern since it holds no persistent data itself.
- **Vercel free (Hobby):** generous bandwidth/build limits for a project this size — not a practical concern here.